Historical House Price Trends Visualizer

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Agenda

01. **Overall Project Goal**

02. **Data**

03. **Data cleaning**

04. **Results**

05. **ARIMA Prediction Model**

Motivation & Goals

The Motivation: Moving Beyond Static Data

Complex Market: Static charts fail to capture 40 years of shifting real estate cycles (1985–2024).

Actionable Context: Homebuyers and investors need to distinguish between historically stable markets and volatile growth.

The Gap: We transform raw HPI (Housing Price Index) data into animations to reveal regional trends that traditional analysis misses.

Project Goals

Dynamic Visualization: Animated state rankings choropleth maps.

Identify Patterns: Answer key questions on the fastest-growing valuations and safest areas to buy.

Future Forecasting: Use ARIMA to project housing trends over the next 10 years.

Data Sources

FHFA Housing Price Index - State Level

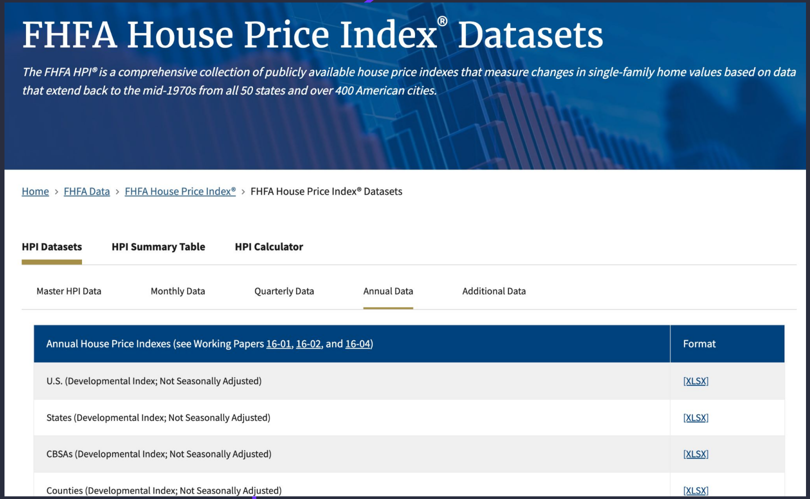
Source: FHFA Annual + Quarterly HPI Datasets

Key fields: State, Abbreviation, FIPS, Year/Quarter, Annual Change (%), HPI

Use: state-level trends, bar chart race, and ARIMA forecasting

Why FHFA?

Long historical coverage of data at both county and state levels. Provides data with consistent geographic identifiers for mapping and comparison



Data Cleaning

Goal: convert raw FHFA HPI files into clean datasets for visualization & forecasting

State Processing:

- Standardize column names & geographic identifiers
- Convert annual change (%) & other numeric fields to numeric types
- Drop rows with missing key fields or invalid values
- Sort records chronologically for consistent analysis
- Group by state and compute 3 year rolling average growth rate
- Preserve cleaned annual growth series for ranking & Forecasting

```
Abbreviation,State,Year,Annual Change (%),Rolling Avg Growth Rate (3yr)
AK,Alaska,1975,,
AK,Alaska,1976,9.22,9.22
AK,Alaska,1977,7.88,8.55
AK,Alaska,1978,10.16,9.086666666666668
AK,Alaska,1979,11.11,9.716666666666667
AK,Alaska,1980,-5.28,5.329999999999999
AK,Alaska,1981,21.16,8.996666666666666
AK,Alaska,1982,24.77,13.549999999999999
AK,Alaska,1983,5.86,17.263333333333332
AK,Alaska,1984,6.78,12.469999999999999
AK,Alaska,1985,-5.5,2.3800000000000003
AK,Alaska,1986,-3.72,-0.8133333333333334
AK,Alaska,1987,-14.88,-8.033333333333333
AK,Alaska,1988,5.49,-4.37
AK,Alaska,1989,-9.89,-6.426666666666667
AK,Alaska,1990,1.62,-0.9266666666666667
AK,Alaska,1991,12.33,1.3533333333333335
```

Choropleth Map Notebook (1)

High-level Overview

A color-coded U.S. map showing year-over-year price changes by state, animated across time.

Abbreviations | State | Year | Annual Change | Rolling Avg Growth (3yr)

Load Data

The dataset contains annual percentage change in the FHFA House Price Index (HPI) for each state, along with a 3-year rolling average to smooth out short-term volatility.

```
df = pd.read_csv("../output/state_growth_rates.csv")
df.head(10)
```

	Abbreviation	State	Year	Annual Change (%)	Rolling Avg Growth Rate (3yr)
0	AK	Alaska	1975	NaN	NaN
1	AK	Alaska	1976	9.22	9.220000
2	AK	Alaska	1977	7.88	8.550000
3	AK	Alaska	1978	10.16	9.086667
4	AK	Alaska	1979	11.11	9.716667
5	AK	Alaska	1980	-5.28	5.330000
6	AK	Alaska	1981	21.16	8.996667
7	AK	Alaska	1982	24.77	13.550000
8	AK	Alaska	1983	5.86	17.263333
9	AK	Alaska	1984	6.78	12.470000

```python
import plotly.graph_objects as go

# Scrape state centroids from Google's public dataset
tables = pd.read_html('https://developers.google.com/public-data/docs/canonical/states_csv')
state_coords = tables[0]
state_coords.columns = ['state', 'latitude', 'longitude', 'name']

# Merge centroids into house price data
df = df.merge(state_coords[['state', 'latitude', 'longitude']], left_on='Abbreviation', right_on='state')

# Create animated choropleth with year slider
fig = px.choropleth(
    df,
    locations='Abbreviation',
    locationmode='USA-states',
    color='Annual Change (%)',
    color_continuous_scale='RdYlGn',
    range_color=(-20, 20),
    scope='usa',
    animation_frame='Year',
    labels={'Annual Change (%)': 'Annual Change (%)'},
    title='Annual House Price Change by State'
)

# State label trace (uses first year's data for coordinates)
coords = df.drop_duplicates(subset='Abbreviation')
label_trace = go.Scattergeo(
    locationmode='USA-states',
    lon=coords['longitude'],
    lat=coords['latitude'],
    text=coords['Abbreviation'],
    mode='text',
    textfont=dict(size=8, color='white'),
    showlegend=False,
    hoverinfo='skip'
)

# Add labels to the base figure
fig.add_trace(label_trace)

# Add labels to every animation frame so they persist
for frame in fig.frames:
    frame.data = (*frame.data, label_trace)

fig.update_layout(margin={"r": 0, "t": 50, "l": 0, "b": 0})
fig.show()
```

Choropleth Map Notebook (2)

Visualizes historical U.S. housing price data (1975–2024)

The choropleth uses a Red–Yellow–Green color scale (range: -20% to +20%) so that:

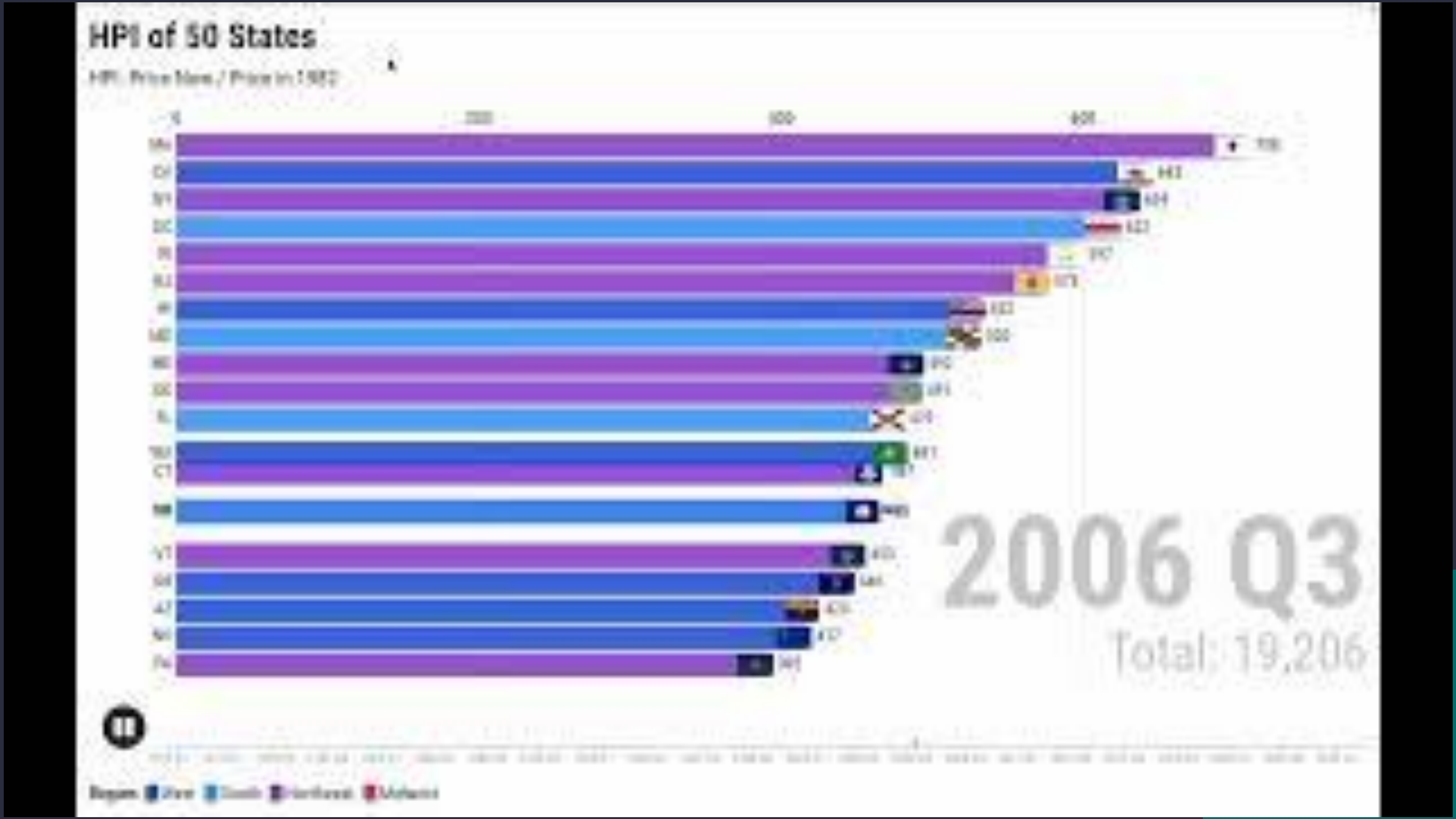
- Red = declining prices (negative annual change)
- Yellow = flat / modest growth
- Green = strong price appreciation

Analysis:

- The 2006–2009 housing crash turning most of the map red
- The post-COVID boom (2020–2022) lighting up nearly every state in green
- Regional differences — coastal states tend to show higher volatility than the Midwest

Bar Chart Race

[Link to Flourish - HPI Race]



Bar Chart Race

High-level Overview

- **Dynamic Leaderboard**: Animated race of HPI rankings for all 50 states from 1975 to 2024.
- **Regional Tracking**: Color-coded regions to identify the dominating regions.

The Transformation Pipeline

- **Engine**: Leveraged **Flourish** to power the interactive Bar Chart Race animation.
- **Reformatting**: from Cleaned HPI files into the layout required for **Flourish.**
- **Enrichment**: Injected state names, regions, and **flag favicons**.

```
Abbreviation,State,Year,Annual Change (%),Rolling Avg Growth Rate (3yr)
AK,Alaska,1975,,
AK,Alaska,1976,9.22,9.22
AK,Alaska,1977,7.88,8.55
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AK,Alaska,1985,-5.5,2.3800000000000003
AK,Alaska,1986,-3.72,-0.8133333333333334
AK,Alaska,1987,-14.88,-8.033333333333333
AK,Alaska,1988,5.49,-4.37
AK,Alaska,1989,-9.89,-6.426666666666667
AK,Alaska,1990,1.62,-0.9266666666666667
AK,Alaska,1991,12.33,1.3533333333333335
```
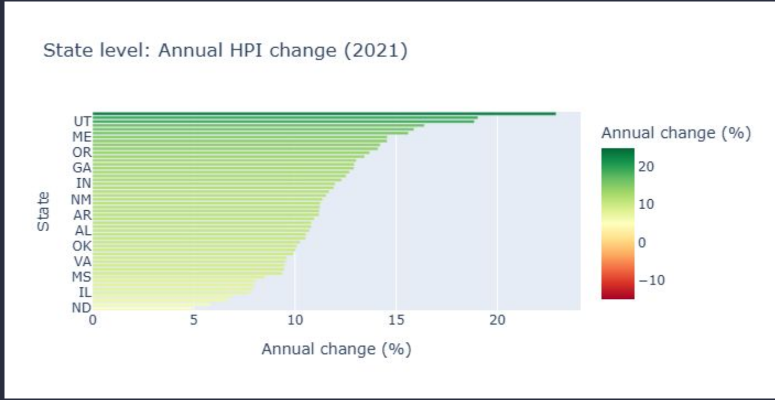
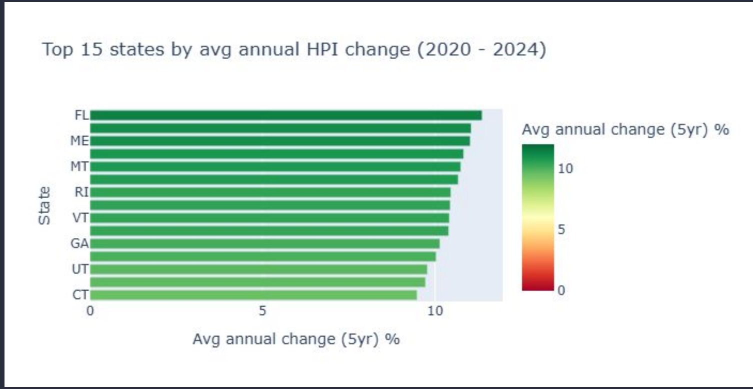

Data | Captions

	A	B	C	D	E	F	G	H	I	J
1	State Name	Region	Image URL	State Abbr	1975 Q1	1975 Q2	1975 Q3	1975 Q4	1976 Q1	1976 Q2
2	Alaska	West	https://flagcdn.com/us-ak.svg	AK	62.03	63.7	66.62	68.27	70.76	70.11
3	Alabama	South	https://flagcdn.com/us-al.svg	AL	72.92	72.09	74.82	71.21	76.87	73.96
4	Arkansas	South	https://flagcdn.com/us-ar.svg	AR	67.17	60.21	64.89	61.98	66.2	68.01
5	Arizona	West	https://flagcdn.com/us-az.svg	AZ	59.47	60.45	59.04	55.95	58.97	62.68
6	California	West	https://flagcdn.com/us-ca.svg	CA	41.67	42.78	44.35	45.75	47.83	50.3
7	Colorado	West	https://flagcdn.com/us-co.svg	CO	54.56	54.58	55.45	54.69	55.21	59.99
8	Connecticut	Northeast	https://flagcdn.com/us-ct.svg	CT	62.85	62.95	61.91	61.84	64.76	63.97
9	District of Columbia	South	https://upload.wikimedia.org/wikipedia/commons/0/(	DC	47.71	46.87	47.3	53.74	53.1	52.84
10	Delaware	Northeast	https://flagcdn.com/us-de.svg	DE	76.84	89.52	86.84	71.35	79.27	85.28
11	Florida	South	https://flagcdn.com/us-fl.svg	FL	66.07	83.59	66.85	68.15	68.14	76.06
12	Georgia	South	https://flagcdn.com/us-ga.svg	GA	73.9	71.68	73.69	73.79	71.98	73.23

Fastest Growth Analysis



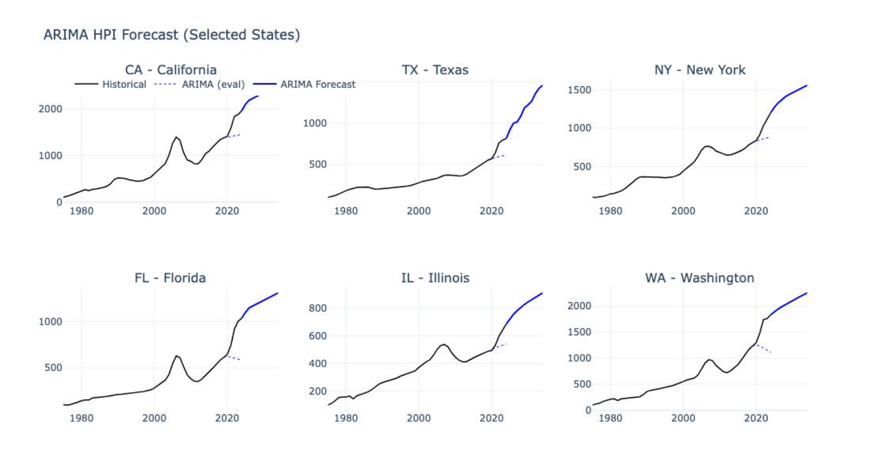


Autoregressive Integrated Moving Average (ARIMA)

Works well for **stable, trend-driven markets** but breaks down when **sudden structural shifts** occur (pandemic housing boom)

15% median error suggests HPI needs multivariate approaches for better accuracy

Short-term forecasts are **usable**, **long-term** are **speculative**



Key Insights

Washington, D.C. has held the highest HPI for a recent 10 years on state level.

Predicted highest HPI state in 2034: **California** with 2500.4

Highest average HPI growth rate from 2020 to 2024: **Florida** with 11.364%

Highest HPI growth rate from 2019 to 2024: **Florida** with 69.3%

Predicted highest HPI growth rate from 2024 to 2034: **Tennessee** with 91.1%

Conclusion

Based on our analysis of the existing HPI datasets and prediction result from the ARIMA model, we can draw following conclusions:

- **Florida** is an excellent place for real estate investment. Over the past five years, it has recorded the highest average growth rate as well as the highest total growth rate. Furthermore, according to the prediction result, it is expected to continue growing steadily over the next ten years.

- **North Dakota** is an ideal choice for purchasing estate, if the primary goal is to keep asset stable during the event of unforeseen circumstances. It is the only state that maintained a positive growth rate during the global economic crisis and the COVID-19 period.

Thank you